



Linear Control Systems 25411

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Assignment 5

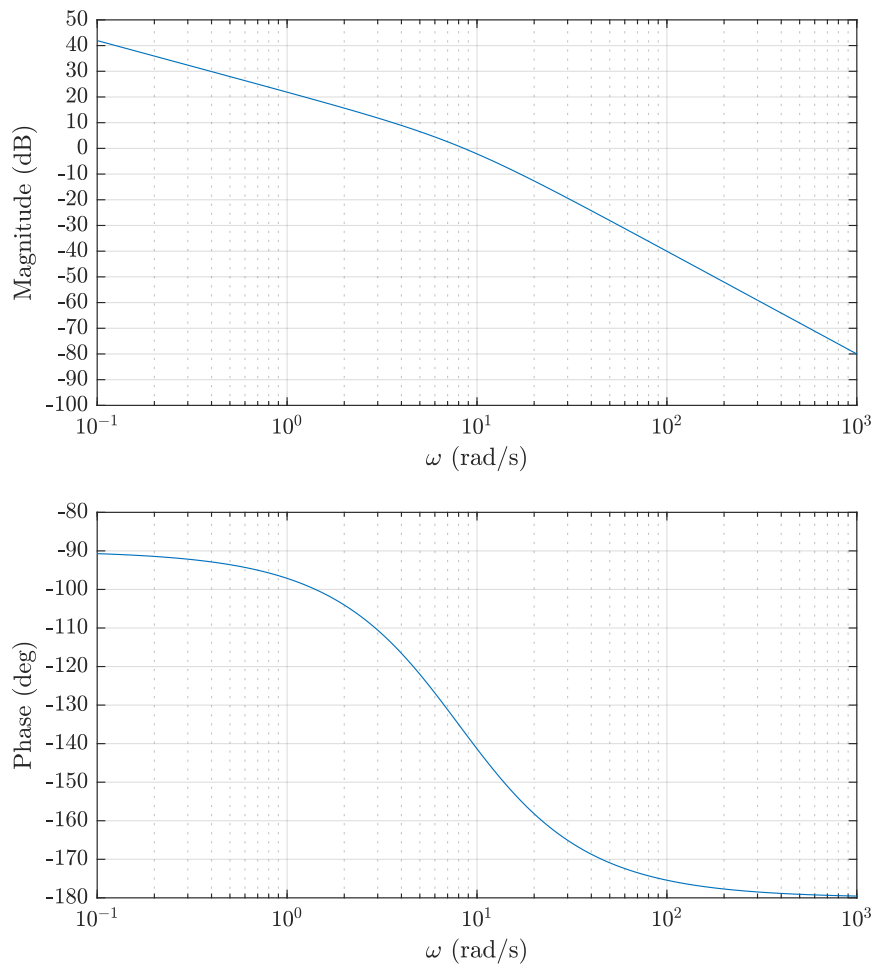
Fall 1404

Due Date: 1404/10/14

Note 1: *purple* problems are bonus ones.

1 Frequency-Domain and Time-Domain Responses

The Bode plots of $G(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$ in a unity feedback control system $T(s) = \frac{G(s)}{1 + G(s)}$ are shown below:



- Find the gain and phase margins and the corresponding crossover frequencies.
- Replace $G(s)$ by $kG(s)$ with $k \geq 1$. Find the range of k for closed-loop stability.
- Replace $G(s)$ by $e^{-\tau s}G(s)$ with $\tau \geq 0$. Find the range of τ for closed-loop stability.
- Find the steady-state error to a unit step input.
- Estimate the percent overshoot, rise time, and settling time of the closed-loop step response.
- Estimate the bandwidth, peak amplitude and peak frequency of the closed-loop frequency response.

2 Stability and l^2 Norm

Consider the set of bounded continuous-time signals. The l^2 norm for this set is defined as follows:

$$\|x\|_2 = \sqrt{\int_0^\infty x^2(t)dt}$$

Using the Parseval's theorem we have:

$$\|x\|_2 = \sqrt{\int_0^\infty x^2(t)dt} = \sqrt{\frac{1}{2\pi} \int_{-\infty}^\infty X(j\omega)X(-j\omega)d\omega}$$

System S is an arbitrary LTI system with transfer function $G(s)$ and $G(s)$ has no poles in the closed right half-plane. $\|S\|$ is defined as follows:

$$\|S\| = \sup_{\|u\|_2 \neq 0} \frac{\|y\|_2}{\|u\|_2}$$

- Prove the following equation:

$$\|y\|_2 \leq \|u\|_2 \max_{\omega} |G(j\omega)|$$

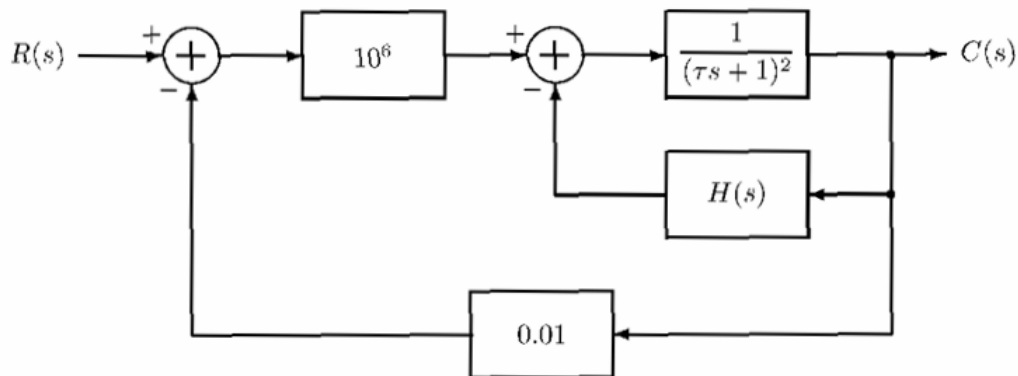
- Prove the following equation:

$$\|S\| = \max_{\omega} |G(j\omega)|$$

Hint: find a signal that satisfies the equality (=) in part a.

3 Minor Loop Compansation

A system with minor loop compansation is shown below, where $\tau = 10^{-4}$.



- a) Do we want $H(s) = \frac{K}{s}$ or $H(s) = Ks$.
- b) Find K such that $\omega_c = 10^6$ rps and $\phi_M = \frac{\pi}{4}$.
- c) What is the dynamic tracking error $|\frac{E}{R}|$ at $\omega = 10^3$ rps?
- d) Your customer wants $\omega_c = 10^6$ rps , $\phi_M = \frac{\pi}{4}$ and $|\frac{E}{R}| < 2 \times 10^{-4}$ at $\omega = 10^3$ rps. choose the $H(s)$ which satisfies all of these criteria'
- $H(s) = \frac{10^{-6}s^2}{10^{-4}s+1}$
 - $H(s) = \frac{10^{-4}s+1}{10^{-6}s^2}$
 - $H(s) = \frac{10^{-5}s^2}{10^{-3}s+1}$
 - $H(s) = \frac{10^{-3}s+1}{10^{-5}s^2}$
 - $H(s) = \frac{10^{-4}s^2}{10^{-2}s+1}$
 - $H(s) = \frac{10^{-2}s+1}{10^{-4}s^2}$
- e) What is $|\frac{E}{R}|$ evaluated at $\omega = 10^3$ rps now?

4 TCP/IP Network Link

The linearized model of a particular network link working under TCP/IP and controlled using a random early detection (RED) algorithm can be described by

$$G(s) = M(s)P(s), \quad H = 1,$$

where

$$M(s) = \frac{0.005L}{s + 0.005}, \quad P(s) = \frac{140625e^{-0.1s}}{(s + 2)(s + 10)}$$

- a) Plot the Nichols chart for $L = 1$. Is the system closed-loop stable?
- b) Find the range of L for closed-loop stability.
- c) Use the Nichols chart to predict %OS and T_s for $L = 0.95$. Make a hand sketch of the expected unit step response.
- d) Verify part c with **step** and **stepinfo** command.
- e) Write MATLAB code to construct the closed-loop transfer function using the **tf** command.
- f) Plot the Bode diagram of closed-loop transfer function and determine the gain margin, phase margin, and crossover frequencies. Verify using the **margin** command.